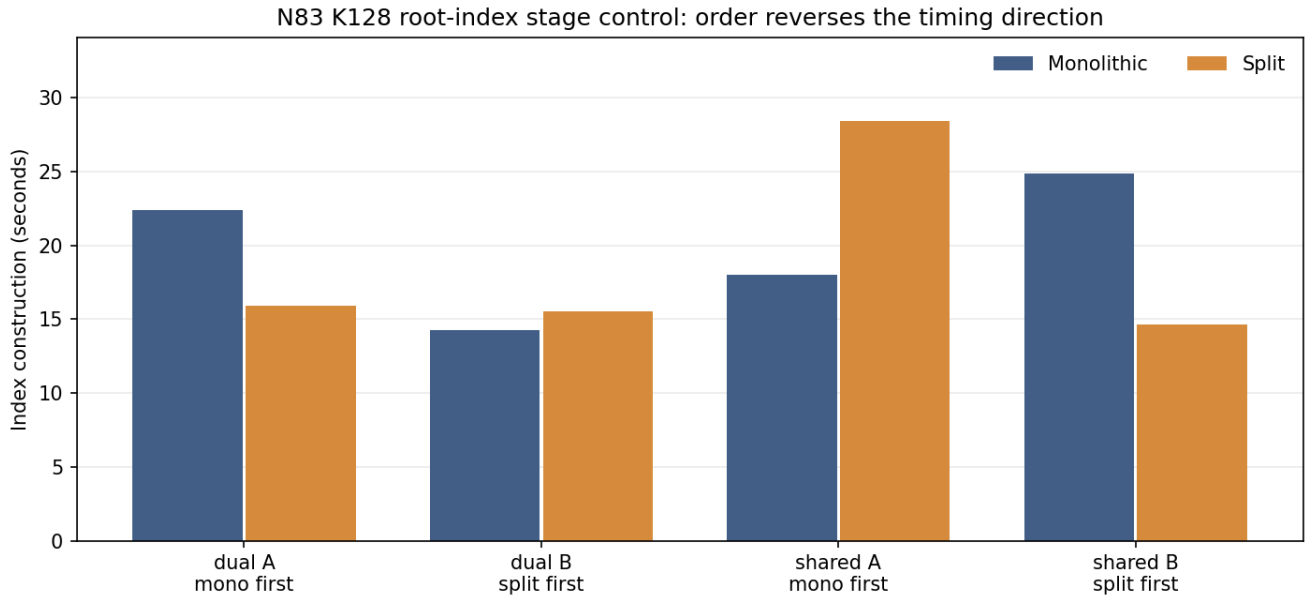
N83 K128 split root index: result

The split exact-root construction preserved the K128 base, state set, and root-entry count. Index-build timing changed direction with run order: the paired local timings do not establish a stage speedup.

At K128 the monolithic five-byte table allocated 10,485,760 bytes and the split tables allocated 13,107,200 bytes. Shared Bloom allocated 2,097,152 bytes, matching monolithic and saving 524,288 bytes compared with separate split-table filters. The frozen K4405 layout formula saves 128 MiB of Bloom space against separate filters and adds 671,088,640 table bytes against monolithic.

Status: STAGE_DIAGNOSTIC_ONLY. Candidate ID and single-target online speedup remain unset. A full-size K4405 same-target scalar replay and Tier 1 paired measurement are the next qualification steps.

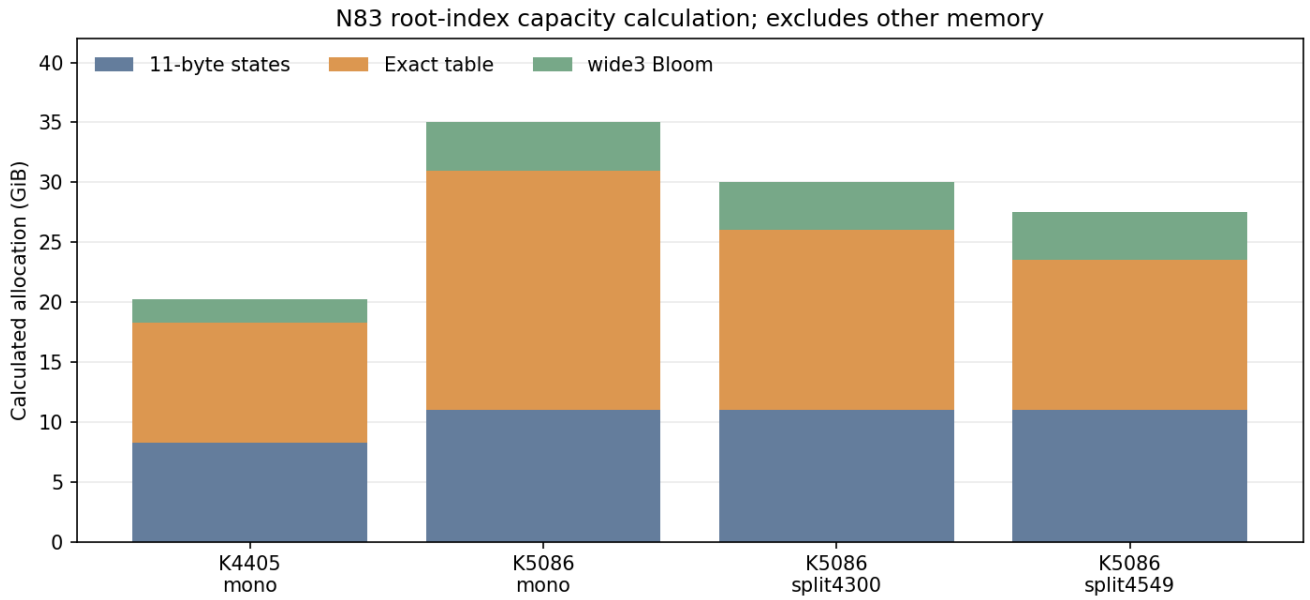
Order	Mode	Mono (s)	Split (s)	Faster stage
k128_a	dual	22.393	15.919	split
k128_b	dual	14.285	15.502	mono
k128_shared_a	shared	18.030	28.390	mono
k128_shared_b	shared	24.862	14.632	split



Source: research/n83_two_tier_root_20261021/runs/{k128_a,k128_b,k128_shared_a,k128_shared_b}/result.json; same study directory contains inputs, stderr, stdout, and runner copies.

Calculated K5086 split capacity

The 30-bit state-ID arithmetic admits K5086 and excludes K5087 and above. With primary-first lookup, K4549 is the unique split point meeting the 2^31 primary and 2^29 extension slot capacities. Five-byte table storage is 12.5 GiB, versus 15 GiB at split K4300 and 20 GiB monolithic. The K5086 point-count formula projects 844,276 points; enumeration and subgroup checks remain to be run.



Calculated arrays only. The 11-byte states, five-byte table, and wide3 union Bloom total 27.50 GiB at K5086/K4549. Other memory and peak RSS are excluded. No K5086 timing or target result is inferred.

Source and proof map: research/n83_two_tier_root_20261021/CAPACITY.md and Capacity.lean; Rust source: examples/koblitz_orbit_dlp_fast.rs.